



AQ7.4: Activity Questions 4 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Consider a function $f: V \times V \rightarrow \mathbb{R}$ where $V \subseteq \mathbb{R}^2$ defined by $f(v, w) = 2v_1 w_1 + 5v_2 w_2$, where $v = (v_1, v_2)$, $w = (w_1, w_2)$. Choose the set of correct options.

☐

f satisfies the symmetry condition of the inner product.

☐

f satisfies the bilinearity condition of the inner product.

☐

f satisfies the positivity condition of the inner product.

☐

f is an inner product.

☐

f is not an inner product.

No, the answer is incorrect.

Score: 0

Accepted Answers:

f satisfies the symmetry condition of the inner product.

f satisfies the bilinearity condition of the inner product.

f satisfies the positivity condition of the inner product.

f is an inner product.

1 point

Consider a function $f: V \times V \rightarrow \mathbb{R}$ where $V \subseteq \mathbb{R}^2$ defined by $f(v, w) = v_1 w_1 - v_1 w_2 - v_2 w_1 + 4v_2 w_2$, where $v = (v_1, v_2)$, $w = (w_1, w_2)$. Choose the set of correct options.

☐

f satisfies the symmetry condition of the inner product.

☐

f satisfies the bilinearity condition of the inner product.

☐

f satisfies the positivity condition of the inner product.

☐

f is an inner product.

☐

f is not an inner product.

No, the answer is incorrect.

Score: 0

Accepted Answers:

f satisfies the symmetry condition of the inner product.

f satisfies the bilinearity condition of the inner product.

f satisfies the positivity condition of the inner product.

f is an inner product.

1 point



Consider two vectors $a = (0.4, 1.3, -2.2)$, $b = (2, 3, -5)$ in R^3 . Choose the set of correct options.

☐

The two vectors satisfy the triangle inequality given by

$$\|a + b\| \leq \|a\| + \|b\|.$$

☐ The two vectors do not satisfy the triangle inequality.

☐

The two vectors satisfy the Cauchy-Schwarz inequality given by

$$|\langle a, b \rangle| \leq \|a\| \|b\|.$$

☐ The two vectors do not satisfy the Cauchy-Schwarz inequality.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The two vectors satisfy the triangle inequality given by

$$\|a + b\| \leq \|a\| + \|b\|.$$

The two vectors satisfy the Cauchy-Schwarz inequality given by

$$|\langle a, b \rangle| \leq \|a\| \|b\|.$$

Level 2

1 point

Consider a function $f: V \times V \rightarrow R$ where $V \subseteq R^2$ defined by

$$f(v, w) = v_1^2 w_1^2 + v_1 w_2^2 + v_2^2 w_1$$

$f(v, w) = v_1^2 w_1^2 + v_1 w_2^2 + v_2^2 w_1$, where $v = (v_1, v_2)$, $w = (w_1, w_2)$. Choose the set of correct options.

☐

f satisfies the symmetry condition of the inner product.

☐

f satisfies the positivity condition of the inner product.

☐

f is an inner product.

☐

f is not an inner product.

No, the answer is incorrect.

Score: 0

Accepted Answers:

f satisfies the symmetry condition of the inner product.

f is not an inner product.

1 point

Consider a vector $a = (a_1, b_1, c_1)$ in R^3 . Which of these is (are) possible candidates for a norm?

☐

$$\sqrt{a_1^2 + b_1^2 + c_1^2}$$

☐
☐

$$a_1 - b_1$$

☐

$$a_1 + b_1 + c_1 \quad a_1 + b_1 + c_1$$



$$\max\{a_1, b_1, c_1\} \max\{a_1, b_1, c_1\}$$



$$\min\{a_1, b_1, c_1\} \min\{a_1, b_1, c_1\}$$



$$\max\{|a_1|, |b_1|, |c_1|\} \max\{|a_1|, |b_1|, |c_1|\}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\sqrt{a_1^2 + b_1^2 + c_1^2} \quad a_1^2 + b_1^2 + c_1^2$$



$$\max\{|a_1|, |b_1|, |c_1|\} \max\{|a_1|, |b_1|, |c_1|\}$$

1 point

Choose the set of correct statements.



$V = \mathbb{R}^2$ and the function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ defined as

$$\langle (x_1, x_2), (y_1, y_2) \rangle = 5x_1 y_1 + 8x_2 y_2 - 6x_1 y_2 - 6x_2 y_1$$

is an inner product on V .



$V = \mathbb{R}^2$ and the function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ defined as

$$\langle (x_1, x_2), (y_1, y_2) \rangle = |x_1| + 2|y_2|$$

is an inner product on V .



$V = \mathbb{R}^2$ and the function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ defined as

$$\langle (x_1, x_2), (y_1, y_2) \rangle = 3x_1 y_1 + 5x_2 y_2$$

is an inner product on V .



$V = \mathbb{R}^2$ and the function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ defined as

$$\langle (x_1, x_2), (y_1, y_2) \rangle = 3x_1 y_1 - 5x_2 y_2$$

is an inner product on V .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V = \mathbb{R}^2$ and the function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ defined as

$$\langle (x_1, x_2), (y_1, y_2) \rangle = 5x_1 y_1 + 8x_2 y_2 - 6x_1 y_2 - 6x_2 y_1$$

is an inner product on V .

$V = \mathbb{R}^2$ and the function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ defined as

$$\langle (x_1, x_2), (y_1, y_2) \rangle = 3x_1 y_1 + 5x_2 y_2$$

is an inner product on V .

[Check Answers](#)

Your score is: 0/6